

Review Articles for searching primary studies:

Sosa-Moscoso, B., Rivadeneira-Limongi, A., Moncayo, F., Loor-Vera, E., Álvarez, D., Vasquez Mena, L. G., Rodas, J. A., & Leon-Rojas, J. E. (2025). Axis I Psychiatric Disorders and Substance Abuse: A Systematic Review of Neuroimaging Findings. *Journal of clinical medicine*, 14(7), 2156.
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Thoma, P., & Daum, I. (2013). Comorbid substance use disorder in schizophrenia: a selective overview of neurobiological and cognitive underpinnings. *Psychiatry and clinical neurosciences*, 67(6), 367–383.
<https://doi.org/10.1111/pcn.12072>

Tesselaar, D. R. M., Schellekens, A. F. A., Homberg, J. R., Booij, J., & Guerrin, C. (2025). Psychiatric comorbidity in substance use disorders, a systematic review of neuro-imaging findings. *Neuroscience and biobehavioral reviews*, 177, 106325. <https://doi.org/10.1016/j.neubiorev.2025.106325>

Smucny, J., & Tregellas, J. R. (2017). Targeting neuronal dysfunction in schizophrenia with nicotine: Evidence from neurophysiology to neuroimaging. *Journal of psychopharmacology (Oxford, England)*, 31(7), 801–811.
<https://doi.org/10.1177/0269881117705071>

Koster, M., Mannsdörfer, L., van der Pluijm, M., de Haan, L., Ziermans, T., van Wingen, G., & Vermeulen, J. (2025). The Association Between Chronic Tobacco Smoking and Brain Alterations in Schizophrenia: A Systematic Review of Magnetic Resonance Imaging Studies. *Schizophrenia bulletin*, 51(3), 608–624. <https://doi.org/10.1093/schbul/sbae088>

Included primary studies:

Cullen, K. R., Wallace, S., Magnotta, V. A., Bockholt, J., Ehrlich, S., Gollub, R. L., Manoach, D. S., Ho, B. C., Clark, V. P., Lauriello, J., Bustillo, J. R., Schulz, S. C., Andreasen, N. C., Calhoun, V. D., Lim, K. O., & White, T. (2012). Cigarette smoking and white matter microstructure in schizophrenia. *Psychiatry research*, 201(2), 152–158.

<https://doi.org/10.1016/j.psychresns.2011.08.010>

Van Haren, N. E., Koolschijn, P. C., Cahn, W., Schnack, H. G., Hulshoff Pol, H. E., & Kahn, R. S. (2010). Cigarette smoking and progressive brain volume loss in schizophrenia. *European neuropsychopharmacology : the journal of the European College of Neuropsychopharmacology*, 20(7), 454–458.

<https://doi.org/10.1016/j.euroneuro.2010.02.009>

Jørgensen, K. N., Skjærvø, I., Mørch-Johnsen, L., Haukvik, U. K., Lange, E. H., Melle, I., Andreassen, O. A., & Agartz, I. (2015). Cigarette smoking is associated with thinner cingulate and insular cortices in patients with severe mental illness. *Journal of psychiatry & neuroscience : JPN*, 40(4), 241–249.

<https://doi.org/10.1503/jpn.140163>

Ringin, E., Cropley, V., Zalesky, A., Bruggemann, J., Sundram, S., Weickert, C. S., Weickert, T. W., Bousman, C. A., Pantelis, C., & Van Rheenen, T. E. (2022). The impact of smoking status on cognition and brain morphology in schizophrenia spectrum disorders. *Psychological medicine*, 52(14), 3097–3115.

<https://doi.org/10.1017/S0033291720005152>

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Journal of psychiatric research, 50, 84–91.

<https://doi.org/10.1016/j.jpsychires.2013.12.004>

Tregellas, J. R., Shatti, S., Tanabe, J. L., Martin, L. F., Gibson, L., Wylie, K., & Rojas, D. C. (2007). Gray matter volume differences and the effects of smoking on gray matter in schizophrenia. *Schizophrenia research*, 97(1-3), 242–249.

<https://doi.org/10.1016/j.schres.2007.08.019>

Yokoyama, N., Sasaki, H., Mori, Y., Ono, M., Tsurumi, K., Kawada, R., Matsumoto, Y., Yoshihara, Y., Sugihara, G., Miyata, J., Murai, T., & Takahashi, H. (2018). Additive Effect of Cigarette Smoking on Gray Matter

Abnormalities in Schizophrenia. *Schizophrenia bulletin*, 44(3), 535–541.

<https://doi.org/10.1093/schbul/sbx092>

Zhang, X., Stein, E. A., & Hong, L. E. (2010). Smoking and schizophrenia independently and additively reduce white matter integrity between striatum and frontal cortex. *Biological psychiatry*, 68(7), 674–677.

<https://doi.org/10.1016/j.biopsych.2010.06.018>

Musket, C. W., Rupert, P. E., Kuo, S. S., Roalf, D. R., Prasad, K. M., Wood, J., Gur, R. C., Almasy, L. A., Gur, R. E., Nimgaonkar, V. L., & Pogue-Geile, M. F. (2026). Shared genetic effects among schizophrenia, substance use disorder, and hippocampal volume in a multiplex extended pedigree sample.

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Koster, M., van der Pluijm, M., Fraikin, M., van Wingen, G., van de Giessen, E., de Haan, L., Vermeulen, J., & Ziermans, T. (2025). Tobacco smoking and gray matter volume in individuals at clinical high risk for psychosis: a longitudinal magnetic resonance imaging study. *Biological Psychiatry Global Open Science*, 5(5), 100539. <https://doi.org/10.1016/j.bpsgos.2025.100539>

Welch, K. A., McIntosh, A. M., Job, D. E., Whalley, H. C., Moorhead, T. W., Hall, J., Owens, D. G. C., Lawrie, S. M., & Johnstone, E. C. (2011). The impact of substance use on brain structure in people at high risk of developing schizophrenia. *Schizophrenia Bulletin*, 37(Suppl 5), 1066–1076.

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Stone, J. M., Bhattacharyya, S., Barker, G. J., & McGuire, P. K. (2012). Substance use and regional gray matter volume in individuals at high risk of psychosis. *European Neuropsychopharmacology*, 22(2), 114–122.

<https://doi.org/10.1016/j.euroneuro.2011.06.004>

Qiu, L., Liang, C., Kochunov, P., Hutchison, K. E., Sui, J., Jiang, R., Zhi, D., Vergara, V. M., Yang, X., Zhang, D., Fu, Z., Bustillo, J. R., Qi, S., & Calhoun, V. D. (2024). Associations of alcohol and tobacco use with psychotic, depressive and developmental disorders revealed via multimodal neuroimaging. *Translational Psychiatry*, 14(1), 326.

<https://doi.org/10.1038/s41398-024-03035-2>